Apartment & Roommate Finder

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1. Brief Introduction

A&R Finder (Apartment and Roommate Finder) is a database-driven web application designed to help university students find off-campus housing and compatible roommates. The platform addresses a common challenge in student housing: relevant information is typically scattered across social media, group chats, and informal networks, making it difficult to compare options and identify suitable living arrangements. By centralizing housing listings and roommate profiles into a single system, A&R Finder gives students a more organized and reliable approach to the housing search process.

The application supports three primary user roles: students seeking housing, students seeking roommates, and apartment providers. Students can create profiles, browse apartment listings, and search for potential roommates based on their preferences. Apartment providers can post and manage listings, update availability, and view inquiries from interested students. Administrators oversee the platform and handle content moderation. This role-based structure ensures that each type of user has access to the features most relevant to their needs.

A central component of the system is the roommate matching functionality. Students complete profiles that capture lifestyle preferences such as sleep schedules, cleanliness habits, noise tolerance, budget range, and social tendencies. The system calculates compatibility scores between users based on these factors, helping students identify potential roommates who share similar living expectations. This reduces the guesswork involved in finding a compatible match and helps prevent conflicts that can arise from mismatched expectations.

The platform also supports collaborative housing searches through group functionality. Students can form groups with friends or matched roommates to coordinate their apartment search together. Within these groups, members can share listings, vote on options, and communicate through built-in messaging. This feature reflects how students typically approach housing searches in practice, working together rather than independently.

From a technical standpoint, A&R Finder was built as a full-stack web application using Next.js and React on the frontend, with Convex (and Type serving as the backend and database layer. The database schema follows relational design principles with normalized tables for users, profiles, listings, matches, groups, and related entities. The system implements role-based access control, server-side validation, and pagination to support efficient querying as the platform scales. Additional features include saved listings, reporting tools for moderation, and an admin dashboard for platform oversight.

2. Normalized Relational Schema

Relational schema diagram (UML Format): https://arfinder.vercel.app/db
Code: **4604**

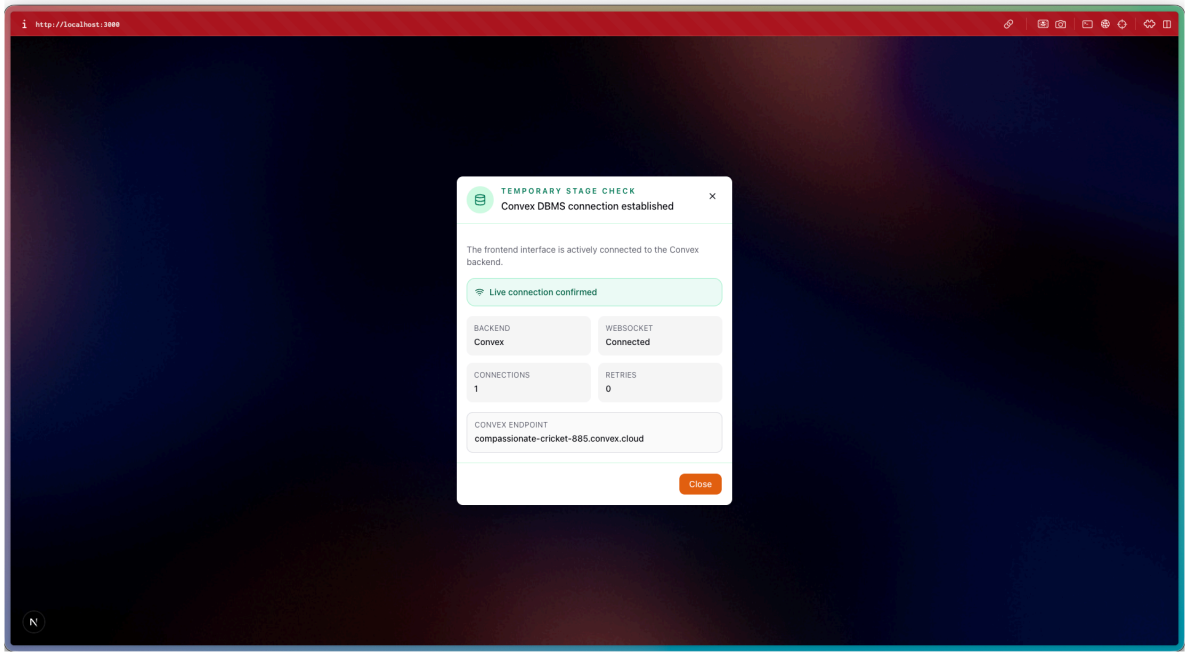
3. All Tables with Sample Data (Major Tables have ≥ 20 tuples, minor tables have fewer)

All tables with sample data: https://arfinder.vercel.app/db
Code: **4604**

(Note: there were too many tables and too much sample data to take a screenshot of each of them, so the above site has all the database information - live)

4. Confirmation of DBMS-Interface Connection

DBMS and Backend platform: Convex (serverless relational-document database)
Programming stack: NextJS, Typescript



6. Roles of Project Members

Daniel Carrillo

Role:

- Contribute to database schema design and refinement, including entity relationships and constraints
- Assist in defining system requirements and translating them into technical specifications
- Design and document value-added system features with a focus on scalability and data integrity
- Support backend development by validating queries, constraints, and data workflows
- Contribute to testing and verification to ensure correctness and consistency across system components

Hursh Soni

Role:

- Assist with DB design and implementation (data models, relationships, etc.)
- Work on frontend design & development
- Assist with backend development
- Work on integration testing
- Project documentation through the semester

Anthony Osei

Role:

- Work on backend development
- Work on project phase documentations
- Assist with database design and implementation

Vivan Verma

Role:

- Work with Anthony to ensure a well-developed backend
- Work with Hursh to also develop a frontend
- Testing

Generative AI

Role:

- Help with frontend features and development
- Generate all the seed data for the major tables with actual personas (too much manual work)
- Help with debugging/identifying issues, especially with the backend
- Help with Convex vs typical relational DB for ERD and schema
- Assist with system architecture and design
- Help understand the project scope and provide ideas
- Clean up tasks, help with documentation, and ERD
- Feedback